

Use of the two-stage neural system in industrial electrical tomography - hybrid approach

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ABSTRACT

This study presents an algorithmic concept that allows obtaining higher-quality tomographic images. The method solves the problem of imaging the interior of industrial tanks, reactors, or pipes. Research focuses on how to solve the inverse problem, which is converting measurements to images. Hybrid tomography combined electrical impedance tomography (EIT) and electrical capacitance tomography (ECT) measurements. The measurement vector was converted into images in two steps. In the first phase, the Long Short-Term Memory (LSTM) neural network was used, thanks to which raw reconstructions were obtained. A second network was then trained to convert the images obtained in the first step into enhanced images. The new method is effective and universal because its use is not limited to one type of tomography.

CCS CONCEPTS

• Computer systems organization; • Software and its engineering; • Computing methodologies;

KEYWORDS

Industrial tomography, Hybrid tomography, Artificial neural networks, Machine learning

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1 INTRODUCTION

The main factors hampering the implementation of electrical impedance tomography (EIT) and capacitance tomography (ECT) installations are the difficulty of obtaining high-resolution images, reconstruction fidelity, and the ability to detect even small inclusions [4]. The quality of electrical tomography imaging depends on many factors, which can be broadly divided into hardware and software [5]. As far as hardware is concerned, the laboratory of Netrix SA has developed an original hybrid tomograph with sets of

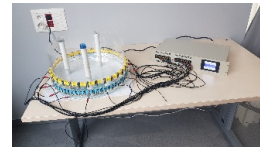


Figure 1: Physical model of the tank for real measurements

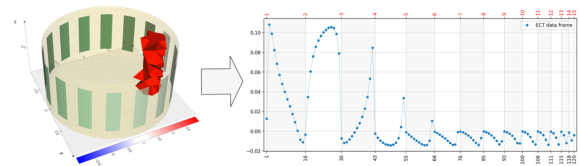


Figure 2: Generating a vector of measurement data by solving the forward problem

electrodes, which enables accurate measurements of both EIT and ECT. A separate problem is converting measurements to images [3]. Tomography of facilities such as reactors, tanks or pipelines is aimed at controlling and monitoring industrial processes. It is a non-invasive method that enables the visualization of crystals or gas bubbles in liquids. Problems of this nature concern the food, chemical, pharmaceutical, refineries, biogas, and other industries. This study presents a two-stage concept of using neural networks to solve the tomographic inverse problem. The method in question makes it possible to obtain a reconstruction of better quality than the standard use of a neural network. A novelty is using the first network to generate a training set for the second network, whose task is to improve the reconstruction.

2 MATERIALS AND METHODS

Fig. 1 shows a physical model of a reservoir reactor with electrodes arranged around it and connected to a tomograph. The following cyber-physical system was used to validate the new method. The measurement vector included 120 ECT and 96 EIT measurements for a total of 216 measurements. 30,000 learning cases were used to train the neural network. They were simulated using the finite element method using the Eiders toolbox and Matlab software [1]. The tetrahedron mesh on which the 3D tomographic image is created consists of 20,445 finite elements. When generating training cases, the forward problem was solved. Fig. 2 shows the generation of the ECT measurement vector for a random pattern image.

On the vertical axis of Fig. 2, some values are related to voltages and electric capacities. However, these values are not direct measurements; they are used to decide what units to use. On the

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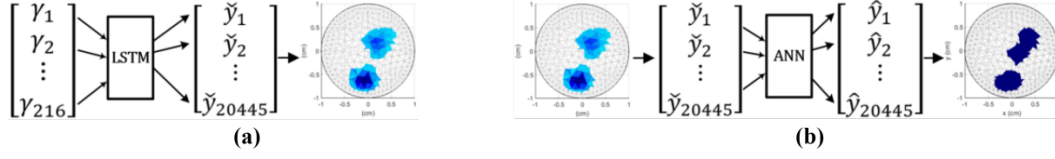


Figure 3: The two-stage process of reconstructing: (a) - the first stage with the LSTM network, (b) - the second stage with the ANN

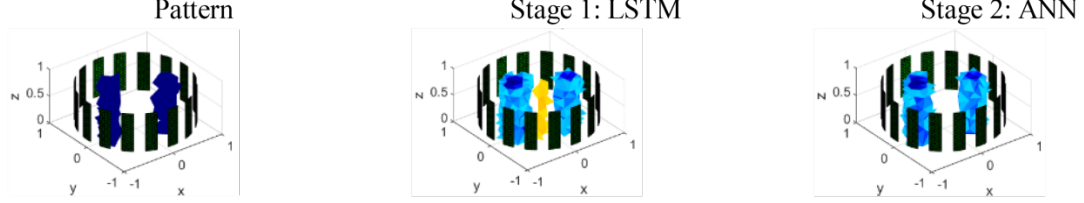


Figure 4: Reconstruction results with the new method: comparing the results after the first and second stage

horizontal axis are individual measurements, of which the ECT is 120 [2].

Fig. 3 shows the two-step process of reconstructing tomographic images. The first step (Fig. 3a) converts the vector 216 of the EIT + ECT measurements into an image with a resolution of 20,445 tetrahedra.

As it can see, the quality of the reconstruction is not perfect. Since the 3D image is created on a tetrahedron mesh, the voxel values are represented as a vector, not a matrix. For this reason, convolutional networks were abandoned. The LSTM network was implemented in the first stage, and in the second stage, it is a shallow artificial neural network (ANN), i.e., a multilayer perceptron. The LSTM network architecture includes 6 layers: sequence input layer [216×1], LSTM layer with 5000 hidden units, Batch normalization, 1D global max pooling, Fully connected with 20,445 neurons, Regression output determined by $Loss = \frac{1}{2L} \sum_{i=1}^L \sum_{j=1}^N (\hat{y}_{ij} - \tilde{y}_{ij})^2$, where L is the

sequence length, N is the number of responses, $\hat{y}_{ij}$ is the target output for responses i,j, and $\tilde{y}_{ij}$ is the reconstructed voxel. The artificial neural network (ANN) architecture in the second stage is as follows: 20445 → ANN(20445) → 20445. Image voxels have binary values. However, both the LSTM network and the ANN were designed as regression models. Our experience has shown that this approach gives better results due to the greater tolerance of errors. Binary values, depending on the applied cut-off filter, all voxels with values outside the tolerance range are visible in the image, distorting it. With regressive voxel values, small errors can be masked by appropriate color calibration.

3 RESULTS AND CONCLUSION

The results of the operation of the presented two-stage model are shown in Fig. 4. The ANN-enhanced image in the second stage is closer to the reference image than the LSTM reconstruction obtained directly from the first stage. The raw reconstructions (LSTM) show a yellow disturbance between the tubes. In Stage 2,

the ANN effectively removes artifacts and improves the shape of the tubes submerged in the water.

The described method focuses on algorithmic issues and involves the use of a two-stage process of transforming measurements into images. In the first step, the LSTM network generates raw reconstructions that serve as training cases to train the ANN. In the second step, the ANN corrects the images obtained in the previous step. The strong point is the combination of EIT and ECT, thus enhancing the measurement vector. The method is intuitively logical, easy to use, works well, and can be used in any tomography. It can be used for both electrical and other kinds of tomography. In the future, work will be done to better align the machine learning models in both stages and use autoencoders to reduce the output vector (image), reducing the deficit of the input number compared to the number of outputs.

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